EXERCISE 03

Step turning

Aim

- To perform step turning operation to produce a cylindrical part of given dimensions.

Required instruments

1. Facing tool
2. Plain turning tool
3. chuck key
4. Tool holder
5. Steel rule
6. Vernier caliper
7. Outside caliper
8. Inside caliper
9. Surface gauge

Tool setting

A facing and a plain turning tool are mounted on the toolpost and their cutting edges are set exactly aligned to the axis of the lathe spindle. The compound rest is set at 0^0 mark.

Holding of the work

1. The metal piece to be machined is held in the chuck of the lathe.
2. Surface gauge is used to check whether the work is held aligning with the lathe axis.

Selection of cutting speed

$$\text{Cutting speed} = \frac{\pi DN}{1000} \text{ m/min.}$$

The above formula is used to arrive at a proper cutting speed for machining and the spindle speed is adjusted suitably.

Facing operation

The lathe is switched on and the facing tool mounted on the toolpost is given feed by the cross-slide perpendicular to the lathe axis. A flat surface on the face of the work is generated.

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